

A Basic L^AT_EX Article

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Abstract

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1 Introduction

The Galton–Watson process is a discrete-time, discrete-space Markov chain. It is a stochastic model in which every member of a population independently either replicates or dies at each generational increment; by these rules, the population fluctuates over the course of its lifetime. Originally formulated by Francis Galton and Henry Watson in the late 19th century, it stands as one of our most fundamental models of stochastic branching.

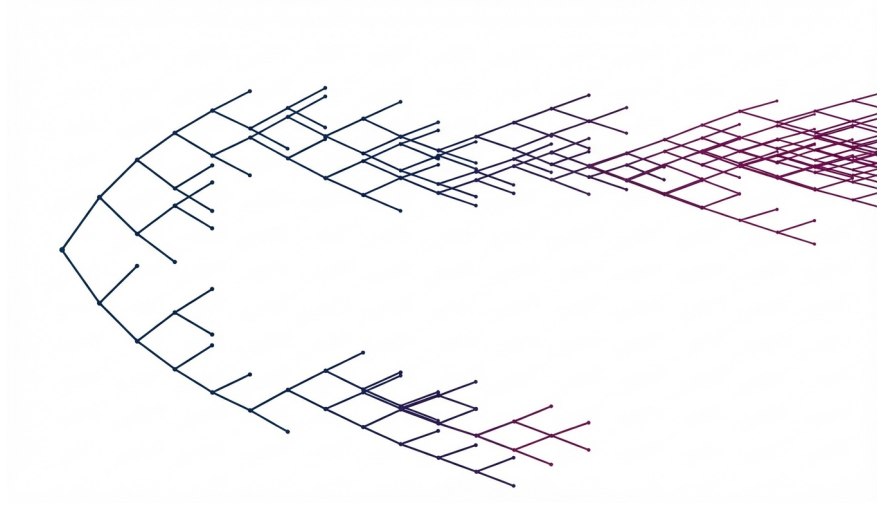
Let Z_n denote the population census at generation n . In the standard case, there is a single initial individual,

$$Z_0 = 1.$$

At the passage to each successive generation n , all Z_{n-1} existing individuals independently

- **replicate** with probability p , or
- **die** with the remaining likelihood $1 - p$.

This is the *simple* Galton Watson process. It is also possible to define the process as having potentially multiple offspring per individual, but here we focus on the classic binary case.



Realisation of the simple Galton-Watson process with $p = 0.58$.

Generally, the expected value of a discrete random variable X is defined by

$$\mathbb{E}(X) = \sum_k \mathbb{P}(X = x_k) x_k,$$

the sum over all possible states x_k in product with their probability of realisation.

Hence, after one generation (with $Z_0 = 1$), we have

$$\mathbb{E}(Z_1) = \mathbb{P}(Z_1 = 2) \cdot 2 + \mathbb{P}(Z_1 = 0) \cdot 0.$$

Replication occurs with probability p , resulting in two offspring. So this simplifies to

$$\mathbb{E}(Z_1) = 2p.$$

By the following argument, we can compute the expected population size at generation n : $\mathbb{E}(Z_n)$.

If we know the value of Z_n , then the conditional expectation of the next generation is

$$\mathbb{E}(Z_{n+1} \mid Z_n) = \sum_{i=1}^{Z_n} \mathbb{E}(Z_1) = 2p Z_n.$$

More generally, if we do not know Z_n , the tower property of conditional expectation gives

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(\mathbb{E}(Z_{n+1} \mid Z_n)) = \mathbb{E}(2p Z_n) = 2p \mathbb{E}(Z_n).$$

Hence,

$$\mathbb{E}(Z_2) = 2p \mathbb{E}(Z_1) = (2p)^2,$$

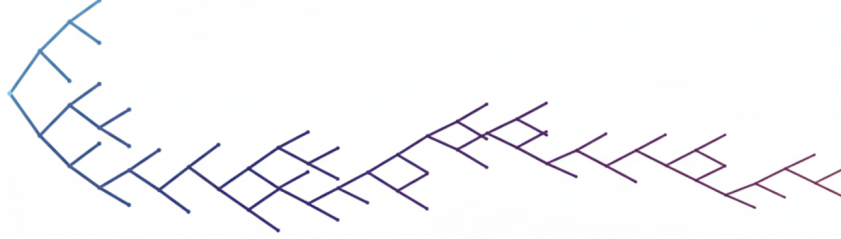
$$\mathbb{E}(Z_3) = 2p \mathbb{E}(Z_2) = (2p)^3,$$

so, induction gives

$$\mathbb{E}(Z_n) = (2p)^n.$$

From this result, we observe a standard behaviour of branching processes:

- If $2p > 1$, then $\mathbb{E}(Z_n)$ grows geometrically with n . In this case, the branching process is said to be *supercritical*.
- If $2p < 1$, then $\mathbb{E}(Z_n)$ decays geometrically with n , and the branching process is *subcritical*.



Subcritical Galton-Watson realisation with $p = 0.47$.

When $2p = 1$ (that is, $p = 0.5$), the Galton–Watson process is said to be *critical*. And in this case, the expectation remains constant:

$$\mathbb{E}(Z_n) = 1^n = 1 \quad \text{for all } n.$$

However, this does *not* mean that the population survives forever. Although the expectation remains positive for all finite generations, as in the supercritical case, when $2p = 1$, ultimate extinction is certain. So the process will eventually fixate at 0, but the expected lifetime of the process is infinite.

And this is a curious mathematical result...

Intuitively, it can be understood by the fact that the total lifetime, T , of this critical branching process is distributed according to a power law.

For such power law distributed variables there is a modestly high chance of extraordinarily large realisations. When you compute a sample mean of a power law distributed variable, the more samples you compute, the larger the mean. It never converges because the occurrence of exceedingly large values pushes the normalised sum higher and higher.

But aside from this intuitive understanding, we will now take an aside to see how this curious result of infinite expected lifetime can be demonstrated.

1.1 Certainty of extinction and expected lifetime for critical GW branching

We will first derive that extinction is certain in the critical case, then we will go on to demonstrate that the expected lifetime of such a process is infinite.

1.1.1 Certainty of extinction

We will see that u_n , the probability of extinction by generation n , can be computed using the probability generating function. And it will follow that the probability of ultimate extinction will

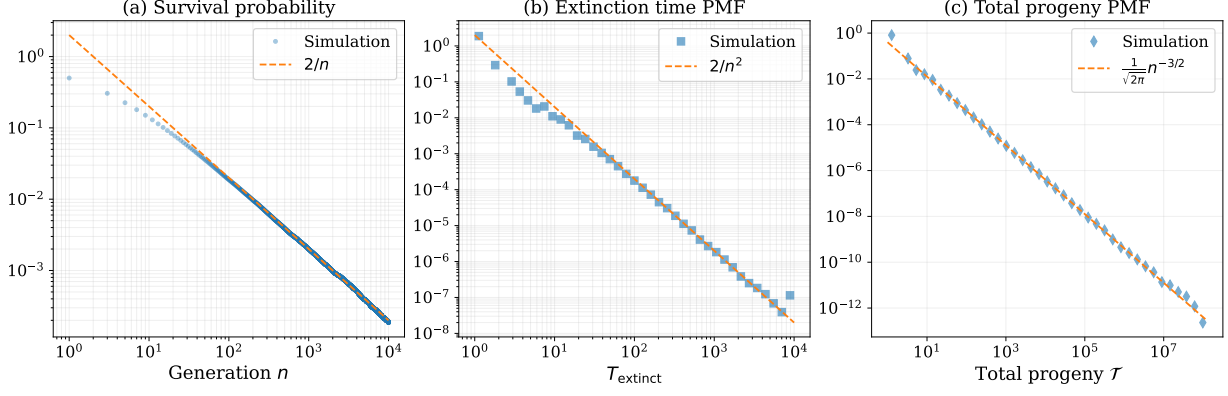


Figure 1: Power-law behavior in a critical Galton–Watson branching process ($p = 0.5$).
(a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late generation asymptotic scaling $S_n \sim 2/n$.
(b) Probability mass function of the extinction time $T_{\text{extinct}} = \inf\{n : Z_n = 0\}$, exhibiting a heavy tail $\mathbb{P}(T_{\text{extinct}} = n) \sim 2/n^2$.
(c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} Z_k$, displaying the classical critical branching asymptotic $\mathbb{P}(\mathcal{T} = n) \sim n^{-3/2}$. (10^6 realizations).

be 1 in both critical and sub-critical branching. $S_n = 1 - u_n$ will denote the survival probability, the likelihood of non-extinction at generation n . A computation using S_n will later demonstrate that the expected lifetime, $\mathbb{E}(T)$, is infinite in the critical case.

The probability generating function is defined by

$$\phi(s) = \mathbb{E}(s^X),$$

for a discrete random variable X .

In our case, the offspring distribution is given by $X = Z_1$, and hence

$$\phi(s) = \mathbb{E}(s^{Z_1}) = ps^2 + (1 - p).$$

For the Galton–Watson processes, it is well known that

$$u_{n+1} = \phi(u_n),$$

where u_n denotes the probability of extinction by generation n .

This relationship can be understood intuitively by considering the fate of the population after the first generation.

- If the process dies in its first step, then extinction has already occurred. In this case, the probability of extinction by any generation is equal to 1. This event occurs with probability $1 - p$.
- If the process replicates in its first step (w/ prob p), then the population reaches generation 1 with two individuals. Extinction by generation $n + 1$ then requires both of these fresh lineages to become extinct by their n th generation; and this happens w/ prob u_n^2 .

Combining these cases yields the recursion

$$u_{n+1} = (1 - p) + pu_n^2 = \phi(u_n).$$

which describes the probability of extinction from one generation to the next. And given the population cannot come back to life once it has died out, u_n will be a non-decreasing sequence bounded above by 1. Conversely, the probability of survival defined by the complement,

$$S_n = \mathbb{P}(Z_n > 0) = 1 - u_n,$$

will be monotonically decreasing and bounded below by 0, but may converge to a positive value $S_\infty \in [0, 1]$. And to determine how these probabilities behave in the late time limit, we can use the fact that they converge and seek steady state solutions.

Substituting $u_n = 1 - S_n$ into equation (1), we obtain

$$1 - S_{n+1} = p(1 - S_n)^2 + (1 - p).$$

Expanding and simplifying gives

$$S_{n+1} = 2pS_n - pS_n^2.$$

To find steady-state solutions, let

$$S_{n+1} = S_n = S_\infty.$$

Then S_∞ satisfies

$$S_\infty = 2pS_\infty - pS_\infty^2,$$

or equivalently,

$$S_\infty(S_\infty + 1 - 2p) = 0.$$

Thus, there are two roots for S_∞ corresponding to two potential steady state solutions.

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2p - 1$$

But since S_n is a probability, it must lie within the interval $[0, 1]$, so when $p < 0.5$, the second root is negative and thus not physically meaningful. Then, when $p > 0.5$, the probability of survival is positive, given by the larger root, $S_\infty = 2p - 1$.

So we have determined that in both the subcritical ($p < 0.5$) and critical ($p = 0.5$) cases, extinction is certain. What about the expected lifetime of the process. How could it be infinite if extinction is certain?

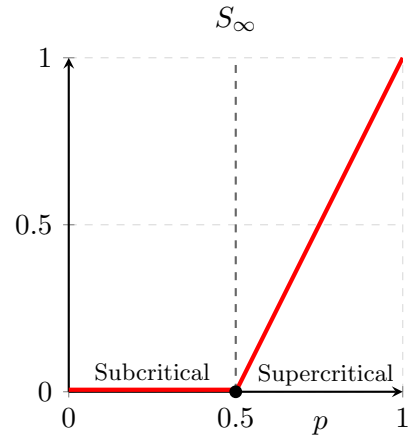
1.1.2 Expected lifetime

We are interested in $\mathbb{E}(T)$ when $p = 0.5$, where T denotes the *extinction time* or the number of generations that the Galton Watson process survives. This is a measure of how many generations the process survives and is defined as

$$T = \inf\{n : Z_n = 0\}.$$

that is, the smallest generation index n at which the population is no longer positive. The process continues over generations

$$0, 1, 2, \dots, T - 1,$$



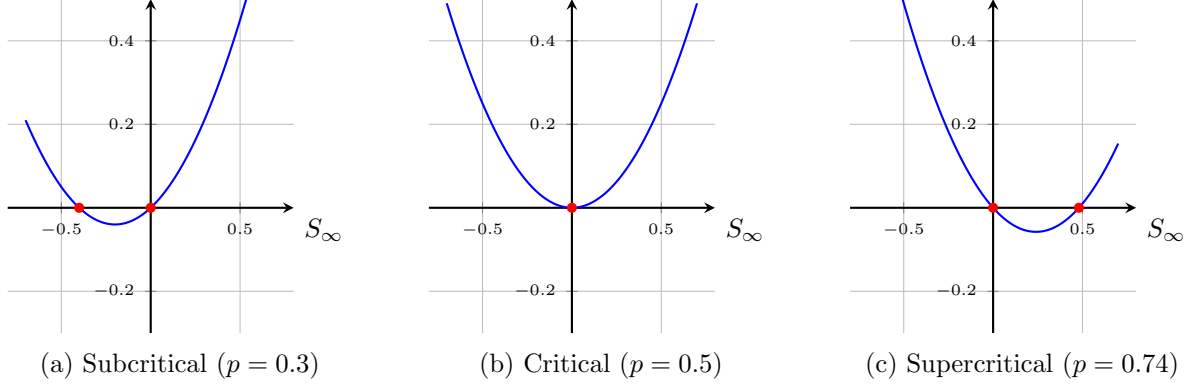


Figure 2: Visualizing $S_\infty(S_\infty + 1 - 2p) = 0$. The roots (red dots) represent equilibrium points. At $p = 0.5$, the roots collide at the origin, and for $p > 0.5$, a stable positive solution $S_\infty = 2p - 1$ emerges.

then fixates to 0, surviving for T generations in total.

To compute the expected time to extinction, we can entertain the following representation.

$$T = \sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}},$$

where $\mathbf{1}_{\{\text{boolean}\}}$ is the indicator function, evaluated:

$$\begin{aligned} \mathbf{1}_{\{\text{true}\}} &= 1 \\ \mathbf{1}_{\{\text{false}\}} &= 0 \end{aligned}$$

The sum involving the indicator function $\mathbf{1}_{\{T > n\}}$ counts one unit for every generation in which the process is alive.

For example, if $T = 5$, then

$$\mathbf{1}_{\{T > n\}} = 1 + 1 + 1 + 1 + 1 + 0 + 0 + \dots$$

Then, to compute the expected value of T ,

$$\mathbb{E}(T) = \mathbb{E}\left(\sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}}\right) = \sum_{n=0}^{\infty} \mathbb{E}(\mathbf{1}_{\{T > n\}}),$$

And given

$$\mathbf{1}_{\{T > n\}} = \begin{cases} 1, & \text{if } T > n, \quad (\text{alive at } n) \\ 0, & \text{otherwise,} \end{cases}$$

with

$$\mathbb{E}(\mathbf{1}_{\{T > n\}}) = \mathbb{P}(T > n).$$

$T > n$ is equivalent to $Z_n > 0$; the process is still alive at generation n . Therefore,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0).$$

In other words, with $S_n = \mathbb{P}(Z_n > 0)$ being the survival probability at generation n ,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n.$$

Here lies an apparent paradox. We have already established that extinction is certain in the critical case, so $S_n \rightarrow 0$ as $n \rightarrow \infty$. Yet the expected time to extinction is infinite, and $\mathbb{E}(T)$ is the sum of S_n which collapses to 0.

Explaining this will require the curious fact that the harmonic series sum is non-convergent.

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

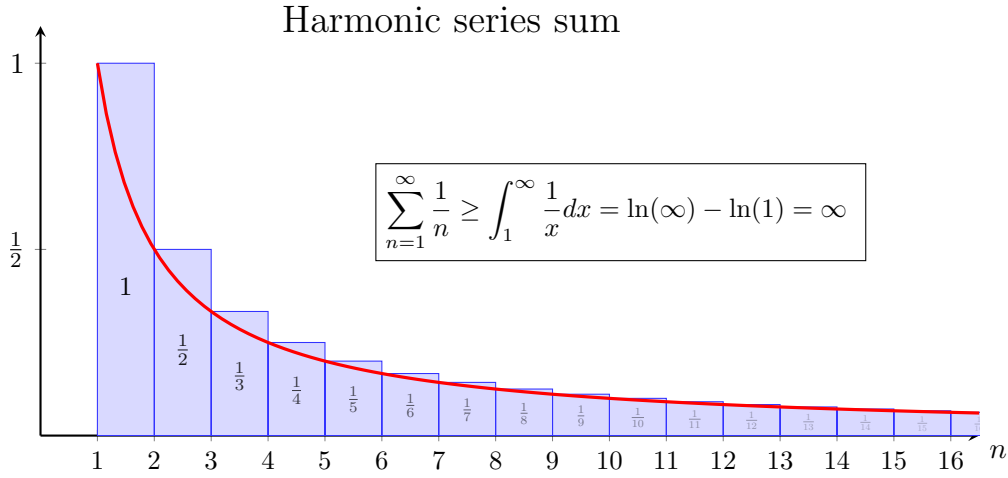


Figure 3: The harmonic series sum diverges. The sum of the areas of the blue rectangles (left) exceeds the area under the red curve (right), which diverges as $x \rightarrow \infty$. Thus, $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$. This fact will be used to show that the expected time to extinction in the critical galton watson process is infinite.

To see how, we first turn our attention to the iterative formula for S_n :

$$S_{n+1} = 2pS_n - pS_n^2.$$

In the critical case, $p = 0.5$, this becomes

$$S_{n+1} = S_n - \frac{1}{2}S_n^2.$$

Rearranging gives

$$S_{n+1} - S_n = -\frac{1}{2}S_n^2.$$

Then, if we consider the late generation limit, $n \gg 1$, the unit 1 can be considered as a diminishingly small increment compared to n , and because of this, the above equation tends to meet the definition of a derivative:

$$\frac{S_{n+1} - S_n}{1} \rightarrow \frac{dS(n)}{dn} = -\frac{1}{2}S(n)^2.$$

So as $n \rightarrow \infty$, the evolution of S_n is increasingly well approximated by this ODE, and thus by its general solution,

$$S(n) = \frac{2}{n + \text{const.}}$$

Hence,

$$S_n \sim \frac{2}{n} \quad \text{as } n \rightarrow \infty, \tag{1}$$

which is supported by the numerical simulation results displayed in Figure 1(a).

Then, coming back to the question of expected lifetime in the critical case, we find that the sum tends towards 2 times the harmonic series in the late generation limit:

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n}.$$

And given S_n is always positive and that the harmonic series sum diverges, it can be concluded that $E(T)$ diverges too. Thus, the expected lifetime of the critical Galton-Watson process is infinite, despite the certainty of extinction.

2 Limiting Conditional Mean $\mathbb{E}(Z_n | Z_n > 0) = 1/A(p)$

With $0 < p \leq 0.5$, the Galton Watson process is certain to cease. But picture this: You initiate a large number of such branching chains with equal $p \leq 0.5$, and watch as they simultaneously progress. Many will die out early, but some may persist for many generations. Here, we will consider the behaviour of those realisations which outlive the majority.

Specifically, we will aim to compute $\mathbb{E}(Z_n | Z_n > 0)$ for $n \rightarrow \infty$, the conditional limiting mean of non-extinct populations, and we will see that it becomes stationary.

To begin, we can decompose the expectation of population size into the two conditional expectations of persistence and cessation:

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot \mathbb{P}(Z_n > 0) + \mathbb{E}(Z_n | Z_n = 0) \cdot \mathbb{P}(Z_n = 0).$$

Since $\mathbb{E}(Z_n | Z_n = 0) = 0$, and $\mathbb{P}(Z_n > 0) = S_n$, this simplifies to

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot S_n.$$

Hence,

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n}.$$

Then, we know that $\mathbb{E}(Z_n) = (2p)^n$, so

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{(2p)^n}{S_n}.$$

And we understand what happens to $(2p)^n$ for large n . But what about S_n ? It may go to 0 in the limit, but how does it get there, and what will this say about the conditional mean?

Recall the recursion

$$S_{n+1} = 2p S_n - p S_n^2.$$

Because $S_n \rightarrow 0$, for growing generation count the quadratic term pS_n^2 diminishes faster than the linear term. Consequently, at large n the recursion is increasingly well approximated by

$$S_{n+1} \sim 2p S_n.$$

And thus it can be shown by induction that this leads to the following.

$$\lim_{n \rightarrow \infty} S_n = A(p) (2p)^n,$$

Where A is a constant dependent on p capturing the cumulative sum of the quadratic terms before they diminish.

Hence,

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}.$$

And it follows that the conditional mean in the late generation limit is given by

$$\lim_{n \rightarrow \infty} \mathbb{E}(Z_n | Z_n > 0) = \lim_{n \rightarrow \infty} \frac{(2p)^n}{S_n} = \frac{1}{A(p)}.$$

So because $A(p)$ is constant, the limiting conditional mean is also constant; it may also be referred to as the quasi-stationary mean population.

But the constant $A(p)$ is still something of a mystery. As we will see, computing an analytic formula is not straightforward.

3 Attempting to compute $A(p)$

We can compute approximate values of $A(p)$ numerically by iterating $S_{n+1} = f(S_n)$ many times to find very near stationary values of S_n at $n \gg 1$. Then, computing $S_{n_{\text{large}}} / (2p)^{n_{\text{large}}}$ gives a value approximating $A(p)$ for that p . Repeating this over a range of probabilities can plot a curve of $A(p)$ for $p \in [0, 0.5)$. And this works reasonably well most of the time. But not when p is from very close to $p = 0.5$, because as $p \rightarrow 0.5$, the required n for S_n to reach near stationarity grows impractically large.

$$\text{Plot of } A(p)$$

There is no known formula which exactly describes $A(p)$, so during the project, Grant and I dedicated efforts to finding one. We tried a number of approaches; for example,

- Evaluating the $S_{n+1} = f(S_n)$ map as an ODE in the large generation limit using $(S_{n+1} - S_n)/1 \rightarrow dS/dn$: $dS/dn = 2pS_n - pS_n^2$.
- A machine learning method of evolving the terms and coefficients of potential formulas using the genetic algorithm.

But unfortunately, nothing gave way to a satisfactory result. It wasn't until months after all but giving up, that I decided to have another look, and found an argument for why $A(p)$ has no closed form solution.

In short, as will be demonstrated below, $A(p)$ can be written in terms of the Koenigs function, $\psi(z)$, which is associated with the discrete logistic map. In the complex plane, the same Koenigs function can be used to plot the well known Julia sets, which are regions of stability and instability

outlined by fractal boundary of infinite roughness. Needless to say, such patterns do not permit closed form description.

Having been discovered in the late 19th century (1884) by French mathematician Gabriel Koenigs, researchers have had some time to understand its properties. And from what I can gather, the Koenigs function, $\psi_r(z)$, which is relevant to our case of the logistic map, $f(z) = rz(1 - z)$, only submits to closed form description in two special cases. Indeed, when $r = 2$ & $r = 4$, $\psi_r(z)$ can be written in terms of a finite number of combined elementary functions:

- $r = 2 \implies \psi_2(z) = -\frac{1}{2}\ln(1 - 2z),$
- $r = 4 \implies \psi_4(z) = (\arcsin(\sqrt{z}))^2.$

These correspond to where in the complex plane, the region of fractal boundary morphs into a circle ($r = 2$) and, the full plane ($r = 4$). But neither of these cases are relevant to our considered range of $p \in [0, 0.5)$ or $r \in [0, 1)$.

So it seems that because $A(p) = f(\psi_{p \in [0, 0.5)}(z))$, $A(p)$ likewise to $\psi(z)$ has no closed form solution.

Before demonstrating more rigorously this fact of no closed formula, we will see something of the attempts made and partial gains both in alaylitically understanding $A(p)$, and approximating it.

3.1 Derivation of $A(p)$ as an infinite product

We may not have found a closed formula for $A(p)$, but here is a derivation of $A(p)$ represented using an infinite product.

Taking, the nonlinear map for S_k ,

$$S_{n+1} = 2pS_n - pS_n^2.$$

To begin, substitute $S_n = 2w_n$,

$$w_{n+1} = pw_n - pw_n^2.$$

This yields the standard logistic form of the logistic map

$$w_{n+1} = rw_n(1 - w_n),$$

($r = 2p$). This is the same logistic map which exhibits the well known behaviour, a characteristic period doubling route to chaos as r increases from 0 to 4.

Now, recall the form of $A(p)$,

$$A(p) = \lim_{n \rightarrow \infty} \left(\frac{S_n}{(2p)^n} \right) = 2 \lim_{n \rightarrow \infty} \left(\frac{w_n}{r^n} \right).$$

Then, because

$$x_n = x_0 \left(\frac{x_1}{x_0} \right) \left(\frac{x_2}{x_1} \right) \cdots \left(\frac{x_{n-1}}{x_{n-2}} \right) \left(\frac{x_n}{x_{n-1}} \right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k},$$

we can represent an infinite limit as an infinite product:

$$\lim_{n \rightarrow \infty} (x_n) = x_0 \prod_{k=0}^{\infty} \left(\frac{x_{k+1}}{x_k} \right).$$

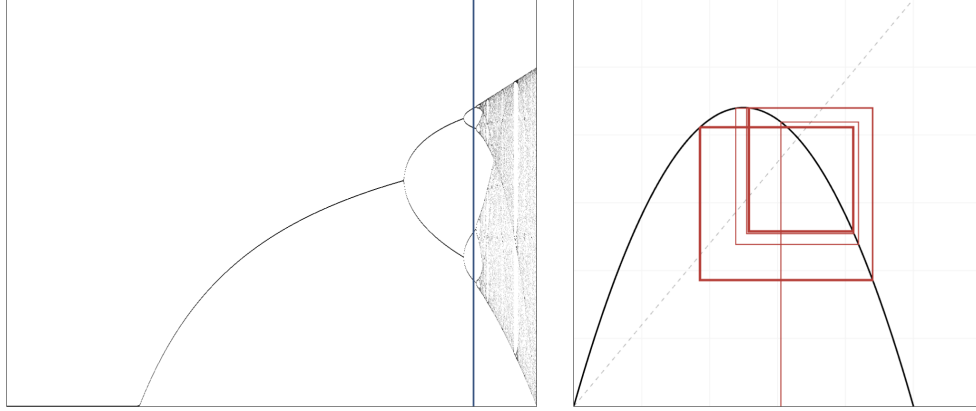


Figure 4: Phase diagram of the logistic map that shows the period doubling of stable states, ultimately resulting in chaotic behaviour. Blue sample line on the left at $r = 3.5255$, right plot shows a trajectory of the map with the same value of r .

With $x_n = w_n/r^n$,

$$A(p) = 2 \lim_{n \rightarrow \infty} (x_n).$$

And converting this to an infinite product as above, we can simplify the fraction,

$$\begin{aligned} \frac{x_{k+1}}{x_k} &= \frac{\frac{w_{k+1}}{r^{k+1}}}{\frac{w_k}{r^k}} = \frac{w_{k+1}}{w_k} \cdot \frac{r^k}{r^{k+1}} \\ &= \frac{rw_k(1 - w_k)}{w_k} \cdot \frac{1}{r} \\ &= 1 - w_k \end{aligned}$$

leading to

$$A(p) = 2 \frac{w_0}{r^0} \prod_{k=0}^{\infty} (1 - w_k)$$

Then, with $w_0 = S_0/2 = 1/2$, and $r^0 = 1$, and pulling out the first ($k = 0$) factor,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k).$$

And finally, reversing the substitution of $S_n = 2w_n$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_n}{2}\right).?? \quad (2)$$

So here we have a different formula for $A(p)$ which can be used to progressively compute an approximation by multiplying successive factors. But this still requires producing the values of S_n by iterating the nonlinear map, so it's not clear that this result brings any material advantage.

3.1.1 Convergence of the product

Here it will be shown that $A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k)$ converges. To know that the product converges, it can be demonstrated to be a necessary and sufficient condition that the following sum converges,

$$\sum_{k=1}^{\infty} w_k.$$

From the recurrence,

$$w_{n+1} = rw_n(1 - w_n),$$

since $w_n > 0$, & $w_n < 1$, it follows that

$$w_{n+1} < rw_n.$$

By induction, $w_n < w_0(r)^n$, and since $0 < r < 1$, w_n decays geometrically, $\sum_{k=1}^{\infty}$ converges, and hence

$$\prod_{k=1}^{\infty} (1 - w_k) \text{ converges.}$$

3.1.2 Advantages of the infinite product formula.

3.2 Derivation of asymptotic formula for $A(p)$ close to $p = 0.5$

Recall the mapping

$$S_n = 2pS_n - pS_n^2, \quad S_0 = 1,$$

And the limiting non-extinct mean which becomes stationary in the limit $n \rightarrow \infty$:

$$\mathbb{E}(Z_n | Z_n > 0) \rightarrow \frac{1}{A(p)} = \frac{(2p)^n}{S_n}.$$

At present, it seems true that $A(p)$ has no closed formula because it can be written in terms of the Koenigs function which is known not to have closed form in the parameter range. So we are left to compute $A(p)$ numerically with $(2p)^n/S_n$ or the product formula (??) by iterating the above mapping, $S_{n+1} = f(S_n)$, to compute a value of S_n at large n . But this method breaks down when p is very close to 0.5 where the process becomes critical. Towards this limit, $A(p) \rightarrow 0$ & $\mathbb{E}(Z_n | Z_n > 0) \rightarrow \frac{1}{A(p)} \rightarrow \infty$, but we can not accurately compute the near run up numerically. And so here we will derive a work around in the form of an asymptotic solution for $A(p)$ close to $p = 0.5$, which can be used to compute $A(p)$ and $1/A(p)$ around this critical point.

Starting with

$$S_n = 2pS_n - pS_n^2,$$

When $p \rightarrow 0.5$, $\epsilon = 1 - 2p \rightarrow 0$, and this value ϵ will be used in the asymptotic analysis.

Rearranging,

$$S_{n+1} - S_n = (1 - 2p)S_n - \frac{1}{2}S_n^2 + \frac{1 - 2p}{2}S_n^2.$$

In the limit as $n \rightarrow \infty$, $S_{n+1} - S_n$ can satisfy the definition of a derivative:

$$\frac{S_{n+1} - S_n}{1} = \frac{dS_n}{dn} = \epsilon S_n - \frac{1}{2}S_n^2 + \frac{\epsilon}{2}S_n^2.$$

We have already established that $S_n \rightarrow 0$ when $n \rightarrow \infty$ for $p \in [0, 0.5)$, and so given $\epsilon \ll 1$, the term $\frac{1}{2}\epsilon S_n^2$ will diminish an order faster than the other two terms,

$$\frac{dS_n}{dn} = \epsilon S_n - \frac{1}{2}S_n^2 + O(\epsilon S_n^2)$$

Hence, for this analysis where we seek a formula of first order $O(\epsilon)$, we can drop the $O(\epsilon S_n^2)$ term.

This leaving us with the ODE,

$$\frac{dS_n}{dn} = \epsilon S_n - \frac{1}{2}S_n^2,$$

(valid for large n and $p \rightarrow 0.5$) which is solvable by separation of variables, giving,

$$S(n) = \frac{2\epsilon C e^{-n\epsilon}}{1 - C e^{-n\epsilon}},$$

where C is a constant of integration. But here we run into what appears at first like a dilemma... Because the ODE is only valid in the $n \gg 1$ limit, we cannot use $S_0 = 1$ to fix the constant of integration.

Briefly, I considered the possibility of using $S_n \rightarrow 0$ and $\epsilon \rightarrow 0$ as $n \rightarrow \infty$, but this just leads to $0 = 0$.

However, there is another way. We can use the fact that in the limit $p \rightarrow 0.5$ ($\epsilon \rightarrow 0$), S_n behaves increasingly like $2/n$ (demonstrated at (1)). Dividing $S(n)$ by $C e^{-n\epsilon}$,

$$S(n) = \frac{2\epsilon}{C^{-1}e^{n\epsilon} - 1}.$$

And because in the case of $x \rightarrow 0$, $e^x = 1 + x + O(x^2)$, it can be seen that with $n\epsilon \ll 0$ (therefore $\epsilon \ll \frac{1}{n}$),

$$n\epsilon = 1 + n\epsilon + O(\epsilon^2).$$

In this case, it follows that

$$\begin{aligned} S(n) &= \frac{2\epsilon}{C^{-1}(1 + n\epsilon) - 1} \\ &\approx \frac{2\epsilon}{(C^{-1} - 1) + C^{-1}n\epsilon} \end{aligned}$$

Then, for $S_n \approx 2/n$, it must be found that the $C^{-1} - 1 = 0$, and so by this, C can be set to 1.

$$S(n) = \frac{2}{n}.$$

Returning to the full formula,

$$S(n) = \frac{2\epsilon e^{-n\epsilon}}{1 - e^{-n\epsilon}},$$

We can now seek the asymptotic form of $\mathbb{E}(Z_n | Z_n > 0)$: In $n \rightarrow \infty$,

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{(2p)^n}{S_n} = \frac{1}{A(p)},$$

And so in $n \rightarrow \infty$ & $\epsilon \rightarrow 0$,

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{(2p)^n}{\left(\frac{2\epsilon e^{-n\epsilon}}{1 - e^{-n\epsilon}}\right)}.$$

But for a moment, consider the following.

$$(2p)^n = (1 - \epsilon)^n = 1 - \epsilon n + O(\epsilon^2)$$

And $e^{-n\epsilon} = 1 - n\epsilon + O(\epsilon^2)$. Hence, at leading order $(2p)^n = e^{-n\epsilon}$ and,

$$\begin{aligned}\mathbb{E}(Z_n | Z_n > 0) &= \frac{e^{-n\epsilon}}{\left(\frac{2\epsilon e^{-n\epsilon}}{1 - e^{-n\epsilon}}\right)} \\ &= \frac{1}{2\epsilon} (1 - e^{-n\epsilon})\end{aligned}$$

And as $n \rightarrow \infty$ (provided $n \gg 1/\epsilon$), this becomes

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{1}{A} \approx \frac{1}{2\epsilon}.$$

Or, viewed another way,

$$A(p) \approx 2\epsilon = 2 - 4p.$$

3.3 Genetic algorithm search attempts

Before finding out that $A(p)$ has no closed formula, I tried a number of methods to find one. And one of those was a Genetic algorithm search based regression. The concept is simple: create 50 functions at random, measure how close each one of them is to correctly describing the curve, then select the best 5 and produce a new population of 50 by creating variations on the best 5. Measure the new 50, and repeat this cycle of variation and selection many times over.

When implementing the first version, I used a search space of combined polynomials; partly because I had a feeling that the solution might show up in that form, and partly because of the relative simplicity of encoding such functions.

In the minimal case, a basic polynomial of the standard form can be used.

$$a_0 + a_1 p + a_2 p^2 + a_3 p^3 + a_4 p^4 + \dots$$

And the "genome" encoding this class of functions would be $[a_0, \dots, a_k]$, to some arbitrarily chosen max order k . The initial 50 genomes get produced as lists of uniformly random generated real coefficients. It turns out that many different particular and substantially varying formulas will produce a curve that looks very close to correct, but we wanted the one exact formula... It occurred to me that the true formula may contain only integer or rational coefficients. So an updated version pursued polynomials of the form,

$$\frac{a_0}{b_0} + \frac{a_1}{b_1} p + \frac{a_2}{b_2} p^2 + \dots + \frac{a_k}{b_k} p^k$$

with $[a_0, \dots, a_k] \in \mathbb{Z}^k$ & $[b_0, \dots, b_k] \in \mathbb{Z}^k$. Then, after some experiments yielding no definite solution, I also tried extending the formula space to

$$\frac{\frac{a_0}{b_0} + \frac{a_1}{b_1} p + \frac{a_2}{b_2} p^2 + \dots + \frac{a_k}{b_k} p^k}{\frac{c_0}{d_0} + \frac{c_1}{d_1} p + \frac{c_2}{d_2} p^2 + \dots + \frac{c_k}{d_k} p^k}$$

with genome $[a_0, \dots, a_k, b_0, \dots, b_k, c_0, \dots, c_k, d_0, \dots, d_k] \in \mathbb{Z}^{4k}$. But after numerous searches, the best that was found merely fit the high precision numerically generated data points quite well. And the formula didn't appear simple enough to represent a probably closed solution:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}$$

It started to become increasingly apparent that a huge number of very different formulas can behave almost identically in a given range.

$\hat{A}_1(p)$ does pass through (0, 0.5) & (0.5, 0) and form a pretty good approximation of the curve visually. Though it does not pass the horizontal axis with the correct gradient of -4 that we know from the asymptotic analysis above. Instead, it passes (0.5, 0) with a gradient of -3.636.

Two years after setting this work aside, I decided to have another go at using an evolutionary regression method like this one to find $A(p)$. It was just starting to become apparent that no formula exists, but I had another go anyway. This time with the option of using AI tools to write a more sophisticated symbolic tree-search program with leveraged time and effort. This new version is capable of searching a space of progressively nested elementary functions of all kinds. Although, of course, a similar conclusion was reached: many functions of varying form will closely resemble the target curve, but no satisfactory simple formula emerged, even on favouring solutions of less component parts.

Below are a couple of generated curves selected arbitrarily from the sea of similarly performing candidates.

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \ln \left(\cos \left(\sqrt{\left| \sqrt{\left| \frac{p/p}{3.19801133824532} \right|} \right|} \right) \right. \\ & \left. + \left(\sin \left(\cos(|p| (0.621865696665606 - p) \cos(e^{e^p})) \right) + e^{\ln(|p|)} \right) p \right) \right) + 0.5982282661889831 \end{aligned}$$

As you can see, there are some redundancies in the formula above such as p/p . These are introduced by the program sort of like neutral mutations in biological evolution having no effect on fitness. Below is the same formula but with these removed.

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \ln \left(\cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right) \right. \\ & \left. + p \left[\sin \left(\cos(|p| (0.621865696665606 - p) \cos(e^{e^p})) \right) + |p| \right] \right) \right) + 0.5982282661889831 \end{aligned}$$

The next one is a bit more concise:

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771$$

PLOTIT

But a big problem with this formula and others produced by this program is that they either do not pass through the horizontal axis at (0.5,0) or they do not have the correct gradient of -4 at the critical point. And these discrepancies cause big inaccuracies when computing the quasistationary mean with $1/A$ as p approaches 0.5 and A draws ever closer to 0. So whilst making the program which plots many Galton-Watson trajectories along with the horizontal red dashed line indicating

the QS mean, I played around with a combined approach seen in $\hat{A}_4(p)$ which aims to correct the near critical behaviour with the asymptotic formula.

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases}$$

This essentially worked, but I found a better solution in a combination of the numerically generated values in combination with the asymptotic $2 - 4p$ in a narrow region of p close to 0.5.

4 Proof of no closed formula for $A(p)$

We know that $A(p)$ cannot be described by a closed formula because it can be written in terms of a Koenigs function which is known not to permit closed form description in the relevant parameter range ($p \in [0, 0.5)$).

4.1 Demonstration of $A(p)$'s relationship to the Koenigs function, $2\psi(1/2)$

We know that $A(p)$ cannot be described by a closed formula because it can be written in terms of a Koenigs function which is known not to permit closed form description in the relevant parameter range ($p \in [0, 0.5)$).

Here will be demonstrated the relationship between $A(p)$ & the Koenigs function. $\psi_r(z)$, associated with the logistic map.

Definition 1. For the logistic map, $f(z) = rz(1 - z)$,

- and fixed point at $z = 0$,
- multiplier $|r| \notin \{0, 1\}$

Gabriel Koenigs proved that there exists a unique analytic function, $\psi_r(z)$, defined such that,

$$\psi(f(z)) = r\psi(z), \quad \& \quad \psi'(0) = 1.$$

In our case, $f(z) = w_{n+1}$, and $z = w_n$.

The Koenigs function linearises the nonlinear mapping:

$$\psi(w_{n+1}) = r\psi(w_n)$$

It will now be shown that $A(p) = 2\psi(w_0) = 2\psi(1/2)$ by demonstrating that

$$\psi(w_0) = \lim_{n \rightarrow \infty} \left(\frac{w_n}{r^n} \right)$$

.

To begin

$$\begin{aligned} \psi(w_1) &= r\psi(w_0), \\ \implies \psi(w_0) &= \frac{\psi(w_1)}{r}, \end{aligned} \tag{3}$$

$$\begin{aligned} \psi(w_2) &= r\psi(w_1), \\ \implies \psi(w_1) &= \frac{\psi(w_2)}{r}. \end{aligned} \tag{4}$$

Then, with (3) & (4),

$$\psi(w_0) = \frac{\psi(w_2)}{r^2},$$

and by induction,

$$\psi(w_0) = \frac{\psi(w_n)}{r^n}. \quad (5)$$

True for any generation, n . And because (5) $\forall n$, we can arbitrarily impose the limit $n \rightarrow \infty$.

$$\psi(w_0) = \lim_{n \rightarrow \infty} \left(\frac{\psi(w_n)}{r^n} \right).$$

And here, there is a trick. Definition 1 states that $\psi'(0) = 1$, meaning that $\lim_{z \rightarrow 0} \psi(z) = z$. Since $w_n = S_n/2 \rightarrow 0$ as $n \rightarrow \infty$,

$$\psi(w_0) = \lim_{n \rightarrow \infty} \left(\frac{w_n}{r^n} \right).$$

Now, recalling that

$$A(p) = 2 \lim_{n \rightarrow \infty} \left(\frac{w_n}{r^n} \right),$$

it follows that

$$A(p) = 2\psi(w_0) = 2\psi(1/2).$$

Remembering that ψ is defined in the context of a particular value of r (or p).

There are a couple of values of the parameter r (2 and 4) for which, as mention above, the Koenigs function does permit a closed form description. Otherwise, and for the whole parameter range $p \in [0, 0.5)$, there is no closed form representation of $\psi_p(1/2)$ in terms of p , and hence neither for $A(p)$.

INSERT THE FRACTAL PLOTS HERE WITGH LINK TO PROGRAM.

4.2 Practically determining $1/A(p)$ as $p \rightarrow 0.5$

In the absence of an exact formula to describe $A(p)$, we are left with approximation and numerical methods. For much of the curve between $p = 0$ and $p = 0.5$, these approaches can give us a computation of the non-extinct branching mean ($1/A$) with very close tolerance, and requiring only a modest number of algorithmic iterations in the numerical case. But there is an issue. As p draws very close to 0.5 corresponding with the near run up to critical branching, $A(p)$ tends to 0 and the nonextict mean $1/A$ tends to infinity. And difernences in A of dininishing magnitude can sway $1/A$ by $O(1)$, which is problematic for the case of an approximate closed formula, but in the case of numerical approximation there is a furthur difficulty. When a sub critical branching process is defined ever nearer to critcality, its lifetime, yet finite, will on the average grow to exceedingly long duration. And in tune with this, its survival probability at generation n , will receed to 0, ever more slowly.

For the numerical apprximation to work, $S_{n+1} = f(S_n)$ must be iterated to a to a sufficiently large n whereby S_n is very close to 0 and all but unchanging from one generation to the next. And at $p \approx 0.5$, this can take more generations than is practical to compute, and hence the approximation begins to fail in that specific range.

For this reason, the follwing regime seems best with present knowledge for computing $A(p)$: use the infinite product apprximation for $0 < p < c$, where c is a number less than but very close to 0.5 (4.9999 for example). Then, for $c < p < 0.5$, use the aysmptotic formula, $A = 2 - 4p$.

$$A(p)_{prox} = \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_n}{2}\right), & \text{when } p < c, \text{ with } N \gg 1 \\ 2 - 4p, & \text{when } p > c \end{cases}$$